

四川大学概率统计（理工）期末考试答案

(2024 — 2025 学年第 1 学期) A 卷

一、填空题(每题3分, 共18分)

1. 1. 2. 1 3. 0.9772
4. 0.8413 5. 0.9198 6. 4.90

二、解答题(7小题, 共82分)

1. (12分) 解. 用 A_i 表示飞机落在第 i 个区域 ($i = 1, 2, 3$), B 表示没有找到飞机. 所求为 $P(A_1|B)$.

已知 $P(A_1) = \frac{1}{2}, P(A_2) = \frac{1}{4}, P(A_3) = \frac{1}{4}, P(B|A_1) = \beta_1, P(B|A_2) = \beta_2, P(B|A_3) = \beta_3$. (4分)

根据贝叶斯公式,

$$P(A_1|B) = \frac{P(A_1)P(B|A_1)}{\sum_{i=1}^3 P(A_i)P(B|A_i)} \quad (4分)$$

$$= \frac{\frac{1}{2}\beta_1}{\frac{1}{2}\beta_1 + \frac{1}{4}\beta_2 + \frac{1}{4}\beta_3} = \frac{2\beta_1}{2\beta_1 + \beta_2 + \beta_3} \quad (4分)$$

2. (12分) 解.

(a) X 和 Y 取值均为 0 或者 1.

$$P(X = 0, Y = 0) = P(U > 2/3) = 1/3, \quad (2分)$$

$$P(X = 1, Y = 1) = P(U \leq 1/3) = 1/3, \quad (2分)$$

$$P(X = 0, Y = 1) = P(U \in (1/3, 2/3]) = 1/3, \quad (2分)$$

$$P(X = 1, Y = 0) = P(\emptyset) = 0. \quad (2分)$$

- (b) $E(X) = 1/3, E(Y) = 2/3, D(X) = 2/9, D(Y) = 2/9, E(XY) = 1/3$.

所以,

$$R(X, Y) = \frac{E(XY) - E(X)E(Y)}{\sqrt{D(X)D(Y)}} = \frac{1}{2}. \quad (4分)$$

3. (10分) 解. X 的密度函数为 $f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad (3分)$

$$E(\sqrt{X}) = \int_0^\infty x^{\frac{1}{2}} e^{-x} dx = \Gamma(\frac{3}{2}) = \frac{1}{2}\Gamma(\frac{1}{2}) = \frac{1}{2}\sqrt{\pi}. \quad (3分)$$

$$\text{而 } E(\sqrt{X}^2) = E(X) = 1. \quad (3分)$$

$$\text{所以, } D(\sqrt{X}) = E(\sqrt{X}^2) - (E(\sqrt{X}))^2 = 1 - \pi/4 \quad (1分).$$

4. (16分) 解.

(a)

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{+\infty} f(x, y) dy \\ &= \begin{cases} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2}{\pi}(x^2 + y^2) dy, & -1 < x < 1; \\ 0, & \text{其他.} \end{cases} \\ &= \begin{cases} \frac{4}{3\pi}(1 + 2x^2)\sqrt{1-x^2}, & -1 < x < 1; \\ 0, & \text{其他.} \end{cases} \end{aligned} \quad (2\text{分})$$

同理, 得

$$f_Y(y) = \begin{cases} \frac{4}{3\pi}(1 + 2y^2)\sqrt{1-y^2}, & -1 < y < 1; \\ 0, & \text{其他.} \end{cases} \quad (2\text{分})$$

可见 $f(x, y) \neq f_X(x)f_Y(y)$, 所以 X 与 Y 不独立. (2分)

(b) 当 $-1 < y < 1$ 时, $f_Y(y) > 0$,

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x, y)}{f_Y(y)} \\ &= \begin{cases} \frac{\frac{2}{\pi}(x^2+y^2)}{\frac{4}{3\pi}(1+2y^2)\sqrt{1-y^2}}, & -\sqrt{1-y^2} < x < \sqrt{1-y^2}; \\ 0, & \text{其他.} \end{cases} \\ &= \begin{cases} \frac{3(x^2+y^2)}{2(1+2y^2)\sqrt{1-y^2}}, & -\sqrt{1-y^2} < x < \sqrt{1-y^2}; \\ 0, & \text{其他.} \end{cases} \quad (2\text{分}) \\ f_{X|Y}\left(x\left|\frac{\sqrt{2}}{2}\right.\right) &= \begin{cases} \frac{3\sqrt{2}}{4}x^2 + \frac{3\sqrt{2}}{8}, & -\frac{\sqrt{2}}{2} < x < \frac{\sqrt{2}}{2}; \\ 0, & \text{其他.} \end{cases} \\ P\left(0 < X < \frac{\sqrt{2}}{4} \middle| Y = \frac{\sqrt{2}}{2}\right) &= \int_0^{\frac{\sqrt{2}}{4}} \left(\frac{3\sqrt{2}}{4}x^2 + \frac{3\sqrt{2}}{8}\right) dx = \frac{13}{64} \quad (2\text{分}) \end{aligned}$$

(c) 先求 Z 的分布函数 $F_Z(z)$. Z 的取值范围是 $[0, 1]$. (2分)

故当 $z < 0$ 时 $F_Z(z) = 0$, 当 $z \geq 1$ 时, $F_Z(z) = 1$.

当 $0 \leq z < 1$ 时,

$$\begin{aligned} F_Z(z) &= P(Z \leq z) = P(X^2 + Y^2 \leq z) \\ &= \iint_{x^2+y^2 \leq z} \frac{2}{\pi}(x^2 + y^2) dx dy \\ &= \frac{2}{\pi} \int_0^{2\pi} d\theta \int_0^{\sqrt{z}} r^3 dr = z^2 \end{aligned} \quad (2\text{分})$$

故 Z 的密度函数为 $f_Z(z) = \begin{cases} 2z, & 0 < z < 1; \\ 0, & \text{其他.} \end{cases} \quad (2\text{分})$

5. (10分) 解.

(a) 检验统计量 $\chi^2 = \frac{(10-1)S^2}{2} = \frac{9}{2}S^2$, (2分)
拒绝域为 $\chi^2 = \frac{9}{2}S^2 > \chi_{1-0.05}^2(10-1) = \chi_{0.95}^2(9) = 16.919$ (4分).

(b) 假设 H_1 成立并且 $\sigma^2 = 4$ 时, $\frac{1}{2}\chi^2 = \frac{(10-1)S^2}{4} \sim \chi^2(9)$. (2分)
所以, 犯第二类错误的概率

$$\begin{aligned}\beta &= P_{\sigma^2=4}(\chi^2 \leq \chi_{0.95}^2(9)) = P_{\sigma^2=4}(\chi^2/2 \leq \chi_{0.95}^2(9)/2) \\ &= F(\chi_{0.95}^2(9)/2) = F(16.919/2) = F(8.4595) = 0.51. \quad (2\text{分})\end{aligned}$$

其中, $F(x)$ 为 $\chi^2(9)$ 的分布函数.

6. (14分) 解.

(a) 因为 $m_1 = E(X) = \theta/2$, 所以, $\theta = 2m_1$.
所以, $\hat{\theta}_M = 2\bar{X}$. (5分)

(b) 似然函数 $L(\theta) = \frac{1}{\theta^n}$, $\theta > \max(X_1, X_2, \dots, X_n)$.
所以, 极大似然估计量为 $\hat{\theta}_L = \max(X_1, X_2, \dots, X_n)$. (5分)

(c) 注意到 $\hat{\theta}_L$ 的取值范围是 $(0, \theta)$. 对于 $x \in (0, \theta)$, $\hat{\theta}_L$ 的分布函数为

$$\begin{aligned}F_L(x) &= P(\max(X_1, X_2, \dots, X_n) \leq x) \\ &= P(X_1 \leq x, X_2 \leq x, \dots, X_n \leq x) \\ &= \prod_{i=1}^n P(X_i \leq x) = \frac{x^n}{\theta^n}\end{aligned}$$

所以, $\hat{\theta}_L$ 的密度函数为 $f_L(x) = \begin{cases} nx^{n-1}/\theta^n, & 0 < x < \theta \\ 0, & \text{其他} \end{cases}$,

于是 $E(\hat{\theta}_L) = \int_{-\infty}^{+\infty} x f_L(x) dx = \frac{n}{n+1}\theta$,

所以 $\hat{\theta}_L$ 不是 θ 的无偏估计量. (4分)

7. (8分) 证明.

$$E(X) = \int_0^{+\infty} \frac{x^{m+1}e^{-x}}{m!} dx = \frac{\Gamma(m+2)}{m!} = m+1 \quad (2\text{分})$$

$$E(X^2) = \int_0^{+\infty} \frac{x^{m+2}e^{-x}}{m!} dx = \frac{\Gamma(m+3)}{m!} = (m+1)(m+2) \quad (2\text{分})$$

$$\begin{aligned}D(X) &= E(X^2) - (E(X))^2 \\ &= (m+1)(m+2) - (m+1)^2 = m+1 \quad (2\text{分})\end{aligned}$$

(或者: 由密度函数知 $X \sim \Gamma(m+1, 1)$, 于是 $E(X) = m+1$, $D(X) = m+1$).
由切比雪夫不等式得:

$$\begin{aligned}P(0 < X < 2(m+1)) &= P(|X - E(X)| < m+1) \\ &\geq 1 - \frac{D(X)}{(m+1)^2} = \frac{m}{m+1} \quad (2\text{分})\end{aligned}$$